

New Generation Power Semiconductor Dynamic Characterization Test System

Introduction

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A question we often ask as a test & measurement equipment manufacturer is 'What is the ideal power semiconductor dynamic test or double pulse test system?'. There are many responses depending on the challenges people encounter at their work. But one of the typical answers for this question is something like this. Insert DUT to the system, push a button and then the system safely performs measurement with excellent accuracy. It is simple to say. However, it is not so easy to make the ideal DPT system.

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Challenges of power device dynamic testing

Conventional challenges associated with power device dynamic test are high voltage and high current. Its high power naturally creates safety issue, measurement accuracy issues for both current and voltage. These are still and continuously the issues to be taken care of when building DPT system. For instance, the test system must be safe to people as the voltage used in the system is significantly higher than 42V which is already dangerous to human. The test resources should also be protected by covering multiple fail scenarios.

Today, the wall of the challenge for the dynamic test becomes even higher due to emerging Wide Band Gap (WBG) power devices such as SiC or GaN.

The supreme material properties of these devices make the switching speed significantly faster than conventional power devices. Although the faster switching speed brings about a lot of benefit such as minimizing the overlap of current and voltage during the transition and making the peripheral component much small, it makes 'high frequency measurement expertise' a mandatory technology when building a dynamic test or DPT system. When the electrical signal in a system goes into high frequency range such as hundreds of MHz or GHz, the entire test system needs to be treated appropriately by considering the effect of parasitic inductance and capacitance. The sudden change of current or voltage, which are often represented as di/dt or dv/dt , couples with surrounding parasitic and causes unexpected voltage or current to appear. They become the trigger of the ringing or reflection in a system that sometimes significantly distorts the measurement waveform [Figure 1]. It easily makes uncontrollable measurement data variability. Even though 'accuracy' is a simple word, you can't get accurate measurement results unless you handle the high frequency signal appropriately. If you build multiple DPT systems, it is common that you don't get the same results across these systems.

Another critical test is temperature dependency measurement. Because WBG devices are used in mission critical application such as automotive, ensuring the switching operation with raised temperature is very important. Hot plate is often used to control the temperature which sometimes heats up not only the DUT but also surrounding circuitry.



Figure 1: Example unexpected ringing seen at the double pulse test

Due to global environmental issue, these WBG devices are intensively studied across the world by government institutions and power semiconductor manufacturers and the performance is continuously and remarkably improved year after year. In other words, the current, voltage, speed and efficiency are concurrently improved, and it will continue moving forward. The test system today won't be applicable a few years later.

Next generation dynamic test technologies and architecture

As discussed in the previous chapter, next generation dynamic testing technologies are necessary to solve today's power device characterization problems. Keysight PD1500A Dynamic Power Device Analyzer/Double Pulse Tester is developed considering these challenges. The first PD1500A made available in April '19 is the initial version of ever evolving DPT system solution series. The architectural concept is 'modular' so that it can keep up with the rapid WBG power device technology advancement going forward.

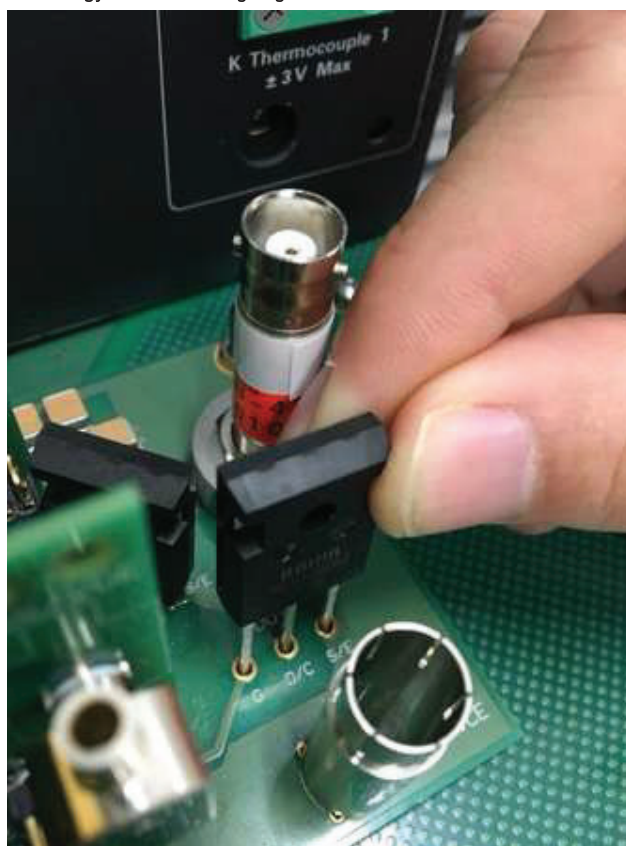


Figure 2: Inserting TO-247 SiC MOSFET into test fixture

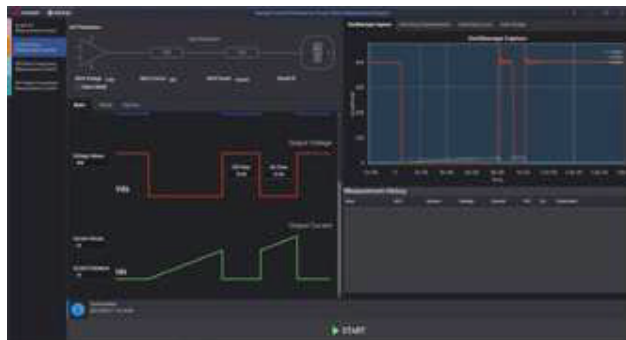


Figure 3: Software GUI and displayed results

The first version focuses on Si and SiC discrete package (TO-247) device. The operation concept is as close as the ideal DPT system discussed in the first section. You can simply insert TO-247 DUT into a test fixture [Figure 2] and close the lid of the safety enclosure. No soldering is necessary. Then, you will adjust parameters on GUI based software followed by pressing 'Start' button. The double pulse test or reverse recovery test is performed automatically and measurement results as well as analyzed data (e.g. extracted parameters) are displayed and stored. [Figure 3] Prior to the measurement, simple calibration process is necessary that allows repeatable and consistent results across multiple test systems.

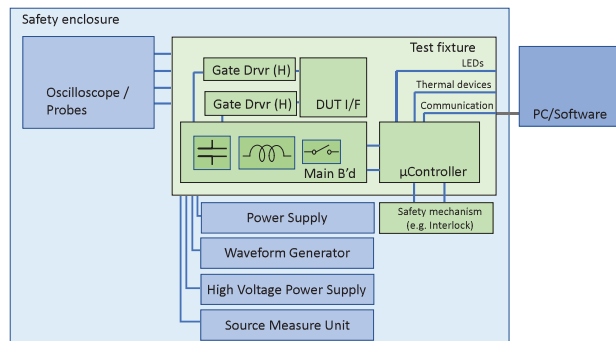


Figure 4: PD1500A block diagram

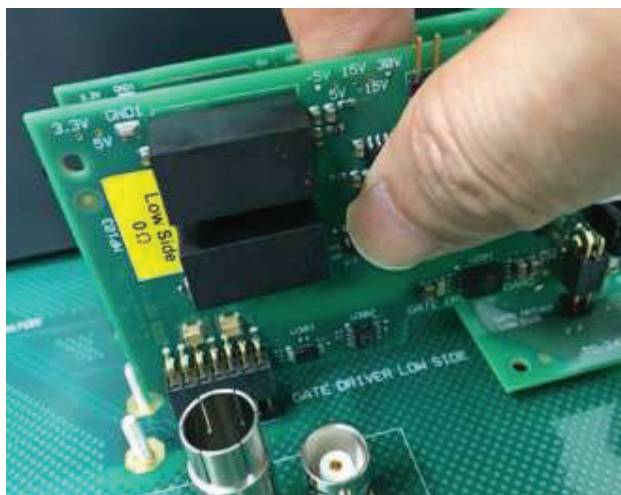


Figure 5: Plug-in type gate driver and DUT I/F

The Figure 4 shows more detailed block diagram. The system is built on modular architecture. For instance, gate driver boards and DUT interface are exchangeable [Figure 5]. It allows easy to change gate resistance to appropriate one by changing gate driver itself. Or, DUT I/F can be changed to SMD type I/F. Even each building block is exchangeable. Therefore, it is only a part of the building blocks to change when upgrading to future technology such as GaN FET or power module with higher current.

The system is also designed to ensure safety operation. The basic safety function such as high voltage detection, interlock, EMO (Emergency Off) switch etc. makes sure the operator safety. The first PD1500A designed to limit the total energy available in the system. Even though maximum voltage and current are 1.2kV and 200A respectively, the user can use the system without worrying about damaging or destroying test resources. Oscilloscope protection also provide additional protection mechanism for test resources.

Temperature dependent measurement can be performed using a small heating device which heats up only the DUT locally. [Figure 6]

For accuracy, the high frequency measurement technologies that are accumulated at Keysight over 80 years are blended into high power measurement technologies. The fast slew rate of the waveform seen

with SiC MOSFET includes high frequency component. If the test circuit has a big stray inductance in the power loop, it acts as a kind of spring that generates ringing. Therefore, careful circuit board design is critical to minimize stray inductance in the power loop. All boards in the PD1500A are designed with the high frequency expertise.

In addition to the calibration techniques applied to each measurement components to produce repeatable and consistent measurement results across multiple test systems, some of high frequency measurement techniques are also applied. For instance, some of components are characterized through s-parameter measurement allowing further sophisticated compensation.

Summary

Keysight PD1500A Dynamic Power Device Analyzer/Double Pulse Tester is the next generation dynamic test platform for power semiconductor device. Various technologies solve challenges seen in the emerging WBG power device testing. The first version allows 'plug-in DUT, push button, and then get results and analysis safely and accurately' operation. Its modular architecture enables further capability expansion as power device/electronics technology advancement with just small change in the system.

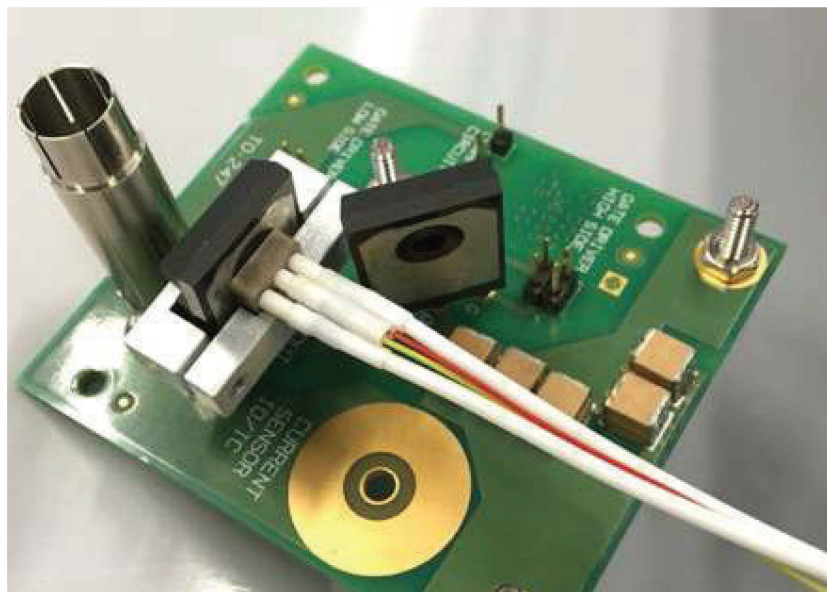


Figure 6: Heating device that locally heats up DUT only

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